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Monitoring Dietary Supplements for Undeclared Diuretics as a Strategy for Preventing Unintentional Doping

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Abstract: Diuretics are essential medicines used in the management of hypertension, heart failure, edema, and other medical conditions. Due to their effectiveness in promoting water loss, many cases of undeclared diuretics in dietary supplements (DSs) have been reported. Most of these DSs were marketed for "weight loss promotion" and "detoxification". Diuretics are regarded as doping compounds as well because of their significant potential to cause weight loss or mask the usage of other doping compounds. The intake of DSs that contain undeclared diuretics can result in serious consequences, including side effects, drug–drug interactions, and the violation of anti-doping rules if such products are taken by professional athletes. The aim of the present study was to develop a rapid, sensitive, and cost-effective high-performance liquid chromatography (HPLC) method for monitoring DSs for undeclared diuretics. The validated method allows for simultaneous monitoring for furosemide, hydrochlorothiazide, and torasemide. The LOD and LOQ were 0.78 μg/mL and 2.38 μg/mL for hydrochlorothiazide, 0.99 μg/mL and 3.01 μg/mL for torasemide, and 0.69 μg/mL and 2.08 μg/mL for furosemide. The method was successfully utilized to screen ten DSs for undeclared diuretics. Contamination was detected in only one sample. These results underscore the critical need for regulatory oversight and independent quality control of DSs. Implementing routine screening could significantly reduce the risk of adverse health effects and unintentional doping among athletes. Ultimately, this study contributes to enhancing consumer safety and ensuring fair competition in sports.

Keywords: inadvertent doping; unintentional doping; dietary supplements; diuretics; doping; food supplements

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1. Introduction

Diuretics are essential medicines used in the management of hypertension, heart failure, edema, and other medical conditions. The intake of diuretics is associated with the promotion of water and electrolyte excretion from the kidneys [1–4]. In general, there are five classes of diuretics: loop diuretics (furosemide, torasemide); potassium-sparing diuretics (spironolactone, triamterene); thiazide diuretics (hydrochlorothiazide); osmotic diuretics (mannitol); and carbonic anhydrase inhibitors (acetazolamide) [5]. The different classes differ in their mechanism of action, molecular site of action, and target site nephron. For example, carbonic anhydrase inhibitors, thiazides, potassium-sparing diuretics, and loop diuretics target the proximal tubule, distal convoluted tubule, late distal tubule, and loop of Henle, respectively [5].

Currently, diuretics are among the most frequently prescribed medications [5]. These drugs must be administrated only after medical examination, and their intake must be discontinued if serious side effects occur. Diuretic-related side effects differ depending on the class of diuretic. In general, these effects include hyponatremia (thiazide diuretics are considered more likely than others to cause it), hypokalemia, hypomagnesemia, impotence, skin rashes, nausea, dizziness, lethargy, etc. [6,7]. For example, thiazide diuretics can increase lithium levels by enhancing lithium reabsorption in the kidney. Studies have reported that thiazide diuretics can cause up to a 40% rise in plasma lithium levels. This condition can lead to tremors, confusion, and potential kidney damage [8,9]. The risk of acute kidney injury is associated with the long-term use of diuretics in combination with nonsteroidal anti-inflammatory drugs (NSAIDs), such as ibuprofen or naproxen [10]. The electrolyte excretion caused by diuretics, especially in cases of hypokalemia and hypomagnesemia, can increase cardiac sensitivity in patients on digoxin therapy and enhance digoxin intoxication. It was established that the risk of hospitalization increases when digoxin is used alongside diuretics [11]. In addition, different drug–drug interactions are reported to occur with diuretics [9–11]. A combination of furosemide and aminoglycosides (for example, gentamicin) can increase the risk of ototoxicity, as aminoglycosides affect cell membranes in the inner ear, leading to increased permeability, which allows furosemide to cause significant damage [12]. Corticosteroids, ethanol, and CNS depressants are also among the drugs that can have adverse effects when combined with diuretics [13].

Although the intake of diuretics must be a result of careful consideration by a physician, the abuse of diuretics has a long history.

Since 1988, diuretics have been regarded as doping agents because of the numerous cases of diuretic abuse by professional athletes. The main key point in the consideration of diuretics as doping agents is their potential to cause water excretion for rapid weight loss and mask the presence of other doping substances. The use of diuretics is banned both in and out of competition, and biological samples of professional athletes are routinely screened for diuretics by the laboratories of the World Anti-Doping Agency (WADA) [6]. However, diuretics currently represent a substantial proportion of all adverse analytical findings reported by WADA laboratories. For example, in 2022, diuretics represented 16% of all adverse analytical findings [14]. Although anabolic agents accounted for the biggest share in this report (42%), the percentage of diuretics was considerable. Often, adverse analytical findings are a result of unintentional doping. However, even in these cases, it is regarded as a violation of the anti-doping rules, and the athletes are considered responsible. In the last decade, unintentional doping has become a serious social health concern that has gained the attention of professional sports and scientific societies [15–17]. Moreover, unintentional doping is considered preventable and should be prevented. Currently, DSs are regarded as one of the gateways to unintentional doping [16]. DSs are widely used among professional athletes and represent an important part of their nutritional regime. Many DSs are cited as products with significant potential to enhance health, athletic performance, or recovery [16]. At the same time, their intake can result in unintentional doping, side effects, drug–drug interactions, and career setbacks. In the last 20 years, many DSs have been reported to be contaminated with prohibited substances [18–23]. Although the teams of many professional athletes carefully choose and monitor which DSs to include in their diets, many athletes turn to unreliable providers of DSs. Moreover, athletes often lack knowledge about DS regulations and the "cons" of the intake of DSs. Currently, in almost all countries in the world, manufacturers of DSs are not required to provide data on the quality analyses of their products [24]. This has resulted in the introduction of many DSs intentionally or unintentionally contaminated with pharmacologically active compounds, including sibutramine, anabolic steroids, and diuretics [18,25–29].

Essential key strategies for the prevention of inadvertent doping are the constant education of athletes and the private monitoring of the quality of DSs before their inclusion in the diet of a specific athlete. Currently, many laboratories provide reliable monitoring of DSs for undeclared compounds.

This work addresses the need for reliable screening methods to minimize the risk of unintentional doping. It offers an effective solution for both regulatory and private laboratories, enhancing safety and transparency in the use of dietary supplements (DSs) for athletes and consumers.

The aim of this study was to develop a simple, sensitive, and cost-effective HPLC method for monitoring DSs for the presence of undeclared diuretics (furosemide, torasemide, and hydrochlorothiazide). The proposed method could be included in programs for monitoring the quality of DSs.

2. Materials and Methods

2.1. Chemicals

The reference standards of hydrochlorothiazide, furosemide, and torasemide HPLC $\geq$ 95% were obtained from Merck KGaA (Darmstadt, Germany). HPLC-grade acetonitrile and formic acid 98–100% were obtained from Merck KGaA (Darmstadt, Germany). Before their use, mobile-phase solvents were subjected to filtration through 0.45 µm membrane filters (Millipore, Milford, MA, USA) and degassed with sonic waves.

2.2. Samples

The analyzed samples included three medications, each containing one diuretic, and ten dietary supplements. The dietary supplements were randomly selected and purchased. All these products were promoted as beneficial for weight loss and detoxification.

2.3. HPLC Analysis

2.3.1. Preparation of Standard Solutions

Stock solutions of hydrochlorothiazide, torasemide, and furosemide were prepared using methanol Merck KGaA (Darmstadt, Germany), each at a concentration of 1 mg/mL. After preparation, the stock solutions were stored at −20 °C until use. Working standard solutions, used for the construction of calibration curves, were freshly prepared by diluting the stock solutions in acetonitrile and water (acidified with 0.1% formic acid) in a 70:30 ratio to obtain mixed working solutions of the three diuretics.

2.3.2. Preparation of Sample Solutions

The medications and dietary supplements were ground and weighed prior to dissolution. One capsule or tablet of each sample was homogenized and then dissolved in 10 mL of methanol. Dissolution was carried out with an ultrasonic bath (Bandelin, Berlin, Germany) for 20 min at 20 °C. Then, the samples were filtered through a 0.45 µm syringe filter and stored at 4 °C until use. The samples were subsequently diluted with acetonitrile and water (acidified with 0.1% formic acid) in a ratio of 70:30 to fall within the calibration range. Triplicates of the samples were made.

2.3.3. Chromatographic Conditions and Instrumentation

The method was developed using a Varian Pro Star HPLC system (Mulgrave, Victoria, Austria) equipped with a UV-VIS detector and software Star Chromatography Workstation Version 6 (Varian Analytical Instruments, Walnut Creek, CA, USA). The column was Supelco Ascentis C18 (4.6 mm × 250 mm, 5 µm), thermostated at 35 °C. The injection

volume was 20 μL, the duration of the method was 12 min, and the wavelength was set at λ = 272 nm. The method was based on a gradient elution mode (Table 1).

Table 1. Mobile-phase composition.

Time (min)	Flow Rate (mL/min)	Water (0.1% Formic Acid)	Acetonitrile
Initial (0)	0.80	75	25
4	0.80	50	50
6	0.80	40	60
8	0.80	30	70
10	0.80	60	40
12	0.80	75	25

3. Results and Discussion

3.1. Method Development

To establish the optimal conditions for the simultaneous determination of the three diuretics using HPLC with UV detection, different mobile phases (MPs) in various ratios were tested: methanol/water, acetonitrile/water, and acetonitrile/water containing formic acid. It was established that the MP composed of acetonitrile/0.1% formic acid in water provided better selectivity than the other options. By optimizing the mobile phase to include 0.1% formic acid, better selectivity and peak resolution were achieved. For the preparation of the standard solutions and test solutions, a mixture of acetonitrile and water (acidified with 0.1% formic acid) was used in a ratio of 70:30. Different flow rates were also tested, and the final flow rate was 0.8 mL/min. The method was performed within 12 min. The retention times were as follows: 5.362 min for hydrochlorothiazide, 6.199 min for torasemide, and 8.518 min for furosemide. The resolution (Rs) was determined to be 4.53 between peaks one and two and 11.59 between peaks two and three.

3.2. Method Validation

The method was validated according to ICH guidelines for linearity, accuracy, precision, limit of detection, limit of quantification, and robustness [30].

3.2.1. Selectivity

Under the optimized chromatographic conditions, complete and excellent baseline separation of all target analytes was achieved. The observed peaks were well resolved for both pure standard solutions and tested DSs samples. The representative chromatograms (Supplementary Materials Figures S1 and S2) demonstrated the absence of interfering peaks at the retention times corresponding to the analyzed drugs.

3.2.2. Linearity

The linearity and range of the method were established with standard solutions of the three diuretics at five different concentrations. The concentrations ranged from 3.125 to 50 μg/mL for the three diuretics. Mixed working solutions were prepared by diluting stock solutions with acetonitrile/0.1% formic acid in water to the required concentrations. The working solutions were injected five times for each concentration, and the corresponding peak area was observed. The linear regression equations and determination coefficients for the three diuretics were as follows: $y = 633050x + 545685$ and $R^2 = 0.9997$ for hydrochlorothiazide, $y = 351280x + 100389$ and $R^2 = 0.9996$ for torasemide, and $y = 587425x + 497605$ and $R^2 = 0.9998$ for furosemide, respectively. Representative peaks are shown in Figure 1.

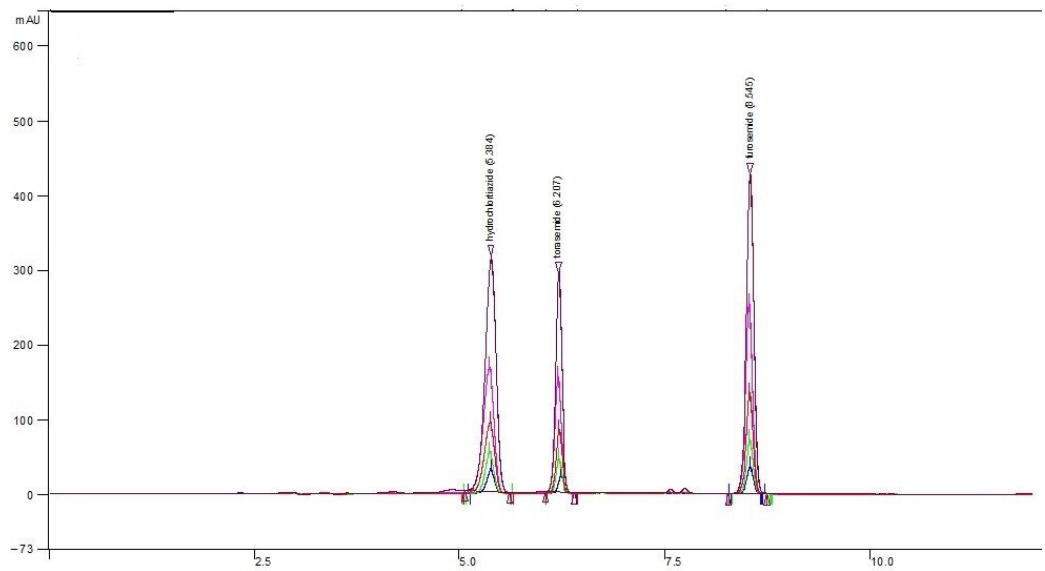

Figure 1. The HPLC chromatogram of standard solutions in the range 3.125–50 µg/mL.

The linearity of the HPLC method was assessed by analyzing a mixture of standard solutions of the three diuretics at different concentrations. The results show excellent linearity for all the compounds analyzed. The determination coefficients (R^2) for the linear regression curves of the three diuretics were above 0.999. This indicates that the method demonstrates a high degree of linearity over the concentration range chosen and is suitable for the quantification of diuretics.

3.2.3. Accuracy

The accuracy was evaluated across the specified range of the analytical method. To verify the accuracy of the developed method, a recovery test was performed at three known concentration levels. The concentration levels were high (25 µg/mL), medium (12.5 µg/mL), and low (6.25 µg/mL). Accuracy is reported as the percentage recovery, calculated by comparing the mean values with the presumed true values (Table 2).

Table 2. The accuracy and precision of the proposed method.

Compound	Concentration µg/mL	Accuracy %	Intra-Day Precision CV (%)	Inter-Day Precision CV (%)
Hydrochlorothiazide	6.25	100.47	0.41	2.04
	12.5	104.43	0.90	1.09
	25	99.81	0.38	0.52
Torasemide	6.25	103.06	1.29	1.97
	12.5	101.26	1.48	1.71
	25	98.00	0.37	1.16
Furosemide	6.25	100.18	0.18	1.24
	12.5	102.99	0.39	0.62
	25	98.99	0.89	1.96

The method's inter-day and intra-day precision was assessed by injecting the same three known concentrations of mixed standard solutions, with each concentration having six replicates. The peaks were observed, and %CVs were calculated. The results of the developed method's precision are presented in Table 2.

To confirm the proposed method's performance, recovery studies were conducted on spiked samples, Sample 1 and Sample 8. Known quantities of the target diuretics—

hydrochlorothiazide, torasemide, and furosemide—were added to these dietary supplement matrices. The concentration levels were high (25 μg/mL), medium (12.5 μg/mL), and low (6.25 μg/mL). The spiked samples were prepared and analyzed in the same way as the unspiked samples. The recovery rates were determined by comparing the spiked samples' concentrations with the theoretical concentrations. The results show recovery rates between 95.5% and 102.7%, thus proving that the method performs well in quantifying the diuretics in complicated matrices. These outcomes substantiate the method's effectiveness, selectivity, and accuracy in detecting undeclared diuretics in supplements.

3.2.4. Detection Limit and Quantification Limit

The detection limit (LOD) and quantification limit (LOQ) were calculated using the standard deviation of the response (σ) and the slope of the calibration curve (S), with the following formulas: LOD = 3.3 σ/S and LOQ = 10 σ/S. The LOD and LOQ were 0.78 μg/mL and 2.38 μg/mL for hydrochlorothiazide, 0.99 μg/mL and 3.01 μg/mL for torasemide, and 0.69 μg/mL and 2.08 μg/mL for furosemide, respectively.

3.2.5. System Suitability

System suitability testing assesses the performance of an analytical system and the effectiveness of separation. During the method's optimization, system suitability parameters were calculated according to the FDA's guidance on the validation of chromatographic methods and the USP's <621> chromatography [31]. Table 3 presents the results obtained under the specified conditions, which meet the acceptable criteria. This confirms that the chromatographic system performs efficiently, yielding well-resolved and symmetric peaks.

Table 3. System suitability parameters using the proposed HPLC method.

Parameter	Hydrochlorothiazide	Torasemide	Furosemide
Capacity factor (K′)	0.623	0.87	1.58
Tailing factor (T)	0.87	1.01	0.90
Number of theoretical plates (N)	7406	33,980	43,388
Selectivity factor (α)	-	1.40	1.82

3.2.6. Robustness

Robustness reflects the reliability of an analytical procedure when small, deliberate changes are made to chromatographic parameters. It was evaluated by varying the column temperature. To assess the effect of temperature variations, a specific temperature range of 32–38 °C was chosen, and the retention time of each compound was assessed. The resolution remained stable despite the changes in temperature. The robustness of the method demonstrates that small temperature variations do not affect the chromatographic separation.

3.3. Method Application

3.3.1. Analyses of Dietary Supplements

The developed and validated method was successfully utilized to monitor the levels of hydrochlorothiazide, torasemide, and furosemide in 10 randomly selected DSs (Table 4). Only one of the analyzed samples was found to contain an undeclared diuretic.

Table 4. Results from analyzed dietary supplements.

Sample №	Country of Origin of the Manufacturer	Place of Purchase	Sample Type	Product Description	Daily Intake Recommendation	Diuretics Detected Per Capsule	Daily Intake
1	Bulgaria	Pharmacy	Capsule	L-Carnitine, Chromium, Green Coffee, Garcinia, Raspberry Ketones	One capsule per day		
2	Bulgaria	Pharmacy	Capsule	L-Taurine, L-Tryptophan, Earth Apple, Bromelain, Field Horsetail, Asian Centella, Coenzyme Q-10, Vitamin B6, Manganese Sulfate, L-Selenomethionine	Two capsules per day		
3	Bulgaria	Online Pharmacy	Capsule	Guarana	Three capsules per day		
4	Bulgaria	Pharmacy	Capsule	Hibiscus, Propolis, Beta-Glucan, D-Manose, Cranberry	Two capsules per day		
5	Bulgaria	Pharmacy	Tablet	*Betula pendula* dry extract	Nine capsules per day		
6	China	Pharmacy	Capsule	Balsam pear, Aloe Vera, *Garcinia Cambodgia*, *Cassia tora*, Hawthorn, Water lily leaves	Two capsules per day	1210 µg *	2420 µg
7	Germany	Pharmacy	Capsule	*Urtica dioica* leaf extr. *Asparagus Officinalis* Extract, Vitamin C, Choline, *Zingiber officinale* Root Extract, Green Tea Extract, Zinc, Vitamin B6, Vitamin B1, Selenium, Vitamin B12	Two capsules per day		
8	Bulgaria	Healthy food shop	Capsule	*Betula pendula* dry extract, *Taraxacum officinale*, Hibiscus, *Equisetum arvense*, Juniperus	One capsule daily		
9	France	Online store	Tablets	*Filipendula ulmaria* L. extr., Caffeine, Guarana, Black radish extract, Artichoke extract, ash leaf extr.	Two capsules per day		
10	Bulgaria	Online Pharmacy	Capsule	*Betula pendula* dry extract	Three capsules per day		

* This value represents the mean of three different replicates.

The diuretic furosemide was found to be present in Sample 6 at a concentration of 1210 µg per capsule. The HPLC chromatogram of the DS containing furosemide is shown in Figure 2. Sample 6 was labeled as a herbal dietary supplement containing cassia tora, hawthorn, and water lily leaves. The product was advertised to promote weight loss. In general, consumers of this product cannot predict or know that it is contaminated with a diuretic.

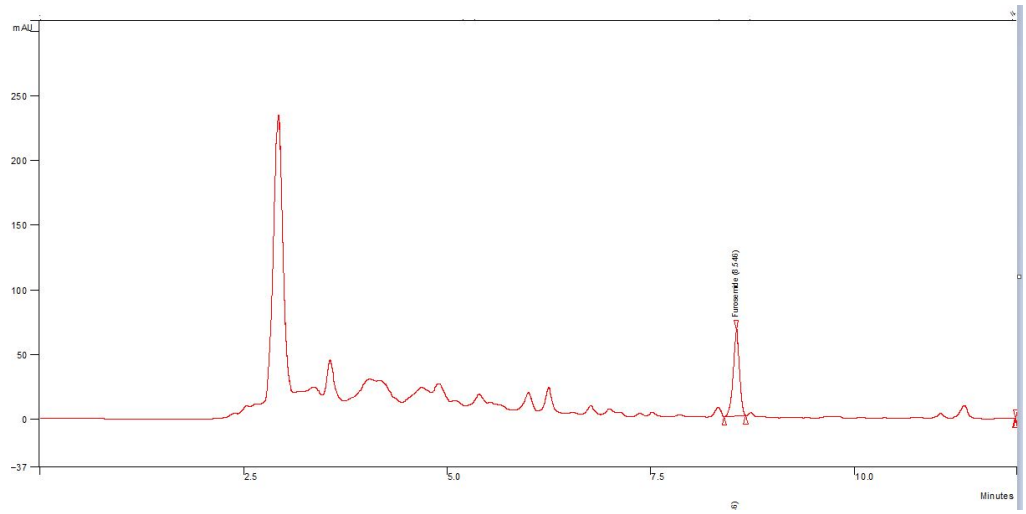

Figure 2. HPLC-UV chromatogram of Sample 6, showing the presence of undeclared DSs.

3.3.2. Analyses of Medicines Containing Diuretics

The tablet forms of the diuretics were diluted to fall within the method's range and were analyzed in three replicates. The quantities of the three evaluated diuretic drug samples are shown in Table 5 (CV% < 3.0%). This assay demonstrated that the method is reliable.

Table 5. Results of the three diuretic drugs.

Diuretic Drugs	Accuracy (%)	CV (%)
Hydrochlorothiazide tablet	97.73	2.34
Torasemide tablet	95.74	0.97
Furosemide tablet	97.83	2.27

Previous studies have confirmed the alarming trend regarding the presence of diuretics in weight-loss DSs. For example, Lee et al. used UHPLC-QOrbitrap to screen 124 DSs and found that approximately 5% were contaminated with diuretics such as hydrochlorothiazide and furosemide. Hydrochlorothiazide and furosemide were detected in concentration ranges of 0.051–76.9 mg/g and 162 mg/g. Their method achieved high sensitivity, with detection limits as low as 0.025 ng/mL and high recovery rates between 81.1% and 115.2% [18]. However, our method strikes a balance between sensitivity and the need for more accessible, cost-effective analysis. Moreira et al. used capillary electrophoresis to analyze laxatives and diuretics in weight-loss DSs, highlighting the potential health risks of such undeclared ingredients. They reported that three of the DSs contained undeclared hydrochlorothiazide [32].

The results of our study align with those of Chen et al., who utilized a quantitative analysis method for the simultaneous detection of multiple diuretics using a single-marker technique. They optimized the efficiency of detecting diuretics in complex mixtures, which is consistent with our goal of enhancing cost-effective yet sensitive detection methods [33]. Similarly, Liu et al. emphasized the importance of developing advanced pretreatment and extraction techniques to improve sensitivity and selectivity in detecting diuretics in various matrices to achieve greater sensitivity and selectivity [34]. Our study's optimized HPLC-UV method leverages a combination of acetonitrile and formic acid to achieve superior peak resolution and reproducibility.

The study by Woo et al. used HPLC and LC-MS/MS techniques to simultaneously analyze 17 diuretics in dietary supplements. First, they used HPLC for the initial screening, followed by LC-MS/MS for confirmation due to its superior sensitivity and specificity.

The method used by Woo et al. is highly robust because it simultaneously analyzes 17 different diuretics. The precision and accuracy were evaluated, with intra-day and inter-day variations kept below 3% for HPLC and below 13% for LC-MS/MS. In comparing recovery rates, after spiking a standard into one dietary supplement, the reported recovery was in the range of 90.4–102.8% [19]. Our method demonstrated recovery values between 95.5% and 102.7% for spiked dietary supplement samples containing hydrochlorothiazide, torasemide, and furosemide.

Detection technologies have recently been improved by increasing both sensitivity and throughput. Goebel et al. developed a rapid screening method using automated solid-phase extraction combined with LC-ESI-MS/MS, specifically for doping control. It is especially useful for the identification of diuretics in urine samples owing to its high sensitivity and ability to manage complex biological environments. Nevertheless, the cost-prohibitive configuration of LC-MS/MS systems limits their practical use in screening DSs [35]. Our HPLC-UV approach, while less sensitive than LC-MS/MS, provides a more accessible alternative for routine quality control in environments with limited resources. This field was further advanced by employing turbulent flow online solid-phase extraction triple quadrupole mass spectrometry for the detection of diuretics and masking agents in urine. Although this approach is extremely sensitive, as it can detect low amounts of prohibited substances in biological samples, its complexity and expense prevent it from being directly used for DS analysis [36]. Stolarczyk et al. developed a sensitive and reliable HPLC-DAD method for the analysis of hydrochlorothiazide, furosemide, torasemide, losartan, valsartan, spironolactone, canrenone, and quinapril, with a run time of 10 min. The analysis was carried out with an LC system (Hitachi, High-Technologies Corp., Tokyo, Japan) on an ACE C18 column (250 × 4.6 mm, 5 µm), with 0.05 M phosphate buffer/acetonitrile/methanol (30:20:50 $v/v/v$) as the mobile phase at a flow rate of 1 mL/min and 25 °C. Despite more advanced instrumentation, they reported higher LOD and LOQ levels for the diuretics analyzed: 9.7 µg/mL and 29.3 µg/mL for hydrochlorothiazide, 3.7 µg/mL and 11.2 µg/mL for furosemide, and 16.6 µg/mL and 50.4 µg/mL for torasemide. Although both methods showed good repeatability, our method outperformed the previous method, with better repeatability for hydrochlorothiazide and furosemide [37]. The HPLC-PDA method developed by Ram et al. was performed on an SGE SS Wakosil II 5C8RS (150 × 4.6 mm × 5 µm) column, using acetonitrile/ammonium acetate buffer (50:50) as the isocratic mobile phase with a flow rate of 1 mL/min. The method was used for the detection of furosemide and spironolactone and demonstrated good precision (99.07–100.58%) and linearity (40–160 µg/mL) for furosemide, as well as a correlation coefficient for furosemide of 0.9953 [38]. Yet another HPLC method developed by Naguib et al. used an LC system (Shimadzu, Kyoto, Japan) on a ZORBAX Eclipse Plus C18 (4.6 × 100 mm × 3.5 µm). The mobile phase was ethanol and deionized water in a volume ratio of 45:55, with pH adjusted to 3.5 using glacial acetic acid. The flow rate was 1 m/min. Three diuretics were analyzed: furosemide, spironolactone, and canrenone. Among these, furosemide is the only diuretic that is also included in our method. The range of furosemide in the referenced method was 5–60 µg/mL, and the LOD and LOQ for furosemide were 1.333 µg/mL and 4.000 µg/mL, respectively [39]. In comparison, our method established a range of 3.125–50 µg/mL for furosemide, with a lower LOD of 0.687 µg/mL and an LOQ of 2.081 µg/mL, which indicates that our method has greater sensitivity for detecting furosemide.

4. Conclusions

In almost all countries in the world, the regulation of DSs could be regarded as quite soft and does not require obligatory analytical control. This has resulted in numerous

cases of poor-quality products contaminated with different pharmacologically active compounds. Moreover, in the last several years, many cases of unintentional doping have been reported after the intake of DSs. Currently, diuretics represent a substantial share of all adverse analytical findings reported by the WADA. The key objective of the present study was the development of a rapid, sensitive, and cost-effective high-performance liquid chromatography (HPLC) method for screening DSs for undeclared diuretics (furosemide, hydrochlorothiazide, and torasemide). The proposed method demonstrates excellent precision, accuracy, and linearity and provides good sensitivity: the LOD and LOQ were 0.69–0.99 μg/mL and 2.08–3.01 μg/mL, respectively, for the different target compounds. The validated HPLC method was used to screen for undeclared diuretics in 10 DSs, all of which were marketed for detoxification/weight loss. One of the samples was found to contain undeclared furosemide (1210 μg per capsule). The recommended daily dose of this DS was two capsules per day, which would provide a daily dose of 2.42 mg of furosemide. This contamination is alarming, as the presence of diuretics or other pharmacologically active agents in DSs poses significant risks, including electrolyte imbalances, dehydration, drug–drug interactions, and unintentional violation of anti-doping regulations. The private and independent monitoring of the quality of DSs before their inclusion in the diets of professional athletes is regarded as a good strategy for the prevention of unintentional doping. Nevertheless, this type of monitoring can be expensive. These analyses should become more accessible and promoted among the professional sports society. The validated HPLC method reported in this paper enables the reliable and fast monitoring of diuretics and, compared to other techniques, is also cost-effective.

Supplementary Materials: The following supporting information can be downloaded at https://www.mdpi.com/article/10.3390/pr13010079/s1, Figure S1: Representative chromatograms of DSs samples (numbers 1–5) and standards; Figure S2: Representative chromatograms of DSs samples (numbers 7–10) and standards.

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